

Title:

Project Summary – Employee Performance Analysis

Results

- Department-wise performance was analyzed successfully.
- Key performance influencing factors were identified.
- Random Forest model was successfully trained and achieved **85% accuracy**

Key Insights

- Employees with higher job satisfaction tend to perform better.
- Work-life balance plays an important role in performance.
- Certain departments like Finance and Sales show comparatively lower performance levels.

Model Performance

- SMOTE improved class distribution.
- Feature importance highlighted top influencing variables.

Recommendations

- Improve employee job satisfaction programs.
- Strengthen work-life balance policies.
- Monitor department-level performance regularly.
- Conduct training programs for low-performing groups.
- Implement reward and recognition systems.

Conclusion

The employee performance dataset was analysed successfully using exploratory data analysis and machine learning techniques.

A Random Forest model was developed to predict employee performance.
SMOTE technique improved class balance and enhanced model performance.

The project provides valuable insights for data-driven decision making.